

Syllabus

Course Title

CEE-200: Advanced Engineering Computation

Academic Department

Chemical and Environmental Engineering

Course Instructors

Markus Petters, Department of Chemical and Environmental Engineering

Web: <https://mdpetters.github.io>

Email: markus.petters@ucr.edu

Office Hours: Friday 3:50-4:50 pm

Course Description

Develops problem-solving techniques for basic engineering systems.

Meeting Time/Location

Lecture: Tu/Th, 12:30 PM - 1:50 PM

Discussion: F, 04:00 PM - 04:50 PM

Location: Riverside Campus | Materials Sci and Engineering | Room 011

Students will need access to a laptop computer. The choice of software is up to the student and may be one of the following: MATLAB, Python, R, or Julia. Students are provided the option to either use a web-based JupyterHub environment or use their personal device for development work.

Software

The choice of programming language is up to the student. Students who choose R can obtain it free of charge from the R Project and the Comprehensive R Archive Network (CRAN).

The R Project for Statistical Computing. <https://www.r-project.org/>. Downloads and packages are distributed through CRAN, <https://cran.r-project.org/>.

Recommended material for getting started with R:

Venables, W. N., Smith, D. M., & the R Core Team (2026). An Introduction to R (version 4.6.1). <https://cran.r-project.org/doc/manuals/r-release/R-intro.html>.

Grolemund, G. (2014). Hands-On Programming with R. Sebastopol, CA: O'Reilly Media. <https://rstudio-education.github.io/hopr/>.

swirl: Learn R, in R. <https://swirlstats.com/>.

Reading

Illowsky, B., & Dean, S. (2023). Introductory Statistics (2nd ed.). Houston, TX: OpenStax, Rice University. <https://openstax.org/details/books/introductory-statistics-2e>.

Bevington, P. R., & Robinson, D. K. (2003). Data Reduction and Error Analysis for the Physical Sciences (3rd ed.). Boston: McGraw-Hill.

Marsden, J. E., & Tromba, A. J. (2012). Vector Calculus (6th ed.). New York: W. H. Freeman. <https://www.macmillanlearning.com/college/us/product/Vector-Calculus/p/1429215089>.

Dawkins, P. (2026). Differential Equations. Paul's Online Notes, Lamar University. <https://tutorial.math.lamar.edu/Classes/DE/DE.aspx>.

Trench, W. F. (2013). Elementary Differential Equations with Boundary Value Problems (free ed.). San Antonio, TX: Trinity University. <https://digitalcommons.trinity.edu/mono/9/>.

Farlow, S. J. (1993). Partial Differential Equations for Scientists and Engineers. New York: Dover Publications. <https://store.doverpublications.com/products/9780486676203>.

Harte, J. (2001). Consider a Cylindrical Cow: More Adventures in Environmental Problem Solving. Sausalito, CA: University Science Books.

Key Learning Outcomes

🔗 Learning Outcomes

1. Become proficient in using R (or similar) for data analysis and simulation.
2. Perform statistical analysis of experimental data, including Monte-Carlo simulation.
3. Solve ordinary differential equations using analytical and numerical methods.
4. Solve partial differential equations using analytical and numerical methods.
5. Perform discrete Fourier analysis on data.

Technology Policy

The use of laptop computers or tablets during lecture is strongly discouraged.

Course Schedule

Tu	Activity	Th	Activity	F	Activity
		Sep 24	Syllabus	Sep 25	No Meeting
Sep 29	Lecture	Oct 1	Lecture	Oct 2	Discussion
Oct 6	Lecture	Oct 8	Lecture	Oct 9	Discussion
Oct 13	Lecture	Oct 15	Lecture	Oct 16	Discussion
Oct 20	Lecture	Oct 22	Lecture	Oct 23	Discussion
Oct 27	Lecture	Oct 29	Lecture	Oct 30	Discussion
Nov 3	Lecture	Nov 5	Lecture	Nov 6	Discussion
Nov 10	Lecture	Nov 12	Lecture	Nov 13	Discussion
Nov 17	Lecture	Nov 19	Lecture	Nov 20	Discussion
Nov 24	Lecture	Nov 26	<i>Thanksgiving/No Class</i>	Nov 27	<i>Thanksgiving</i>
Dec 1	Lecture	Dec 3	Lecture	Dec 4	No Meeting
Dec 7	Final Exam (Mon) 3:00 PM-6:00 PM	Dec 10	No Meeting	Dec 11	No Meeting

Evaluation of Student Performance

The grade components in the class include:

Activity	Weight	Details
Projects	60%	Each module will have one or more graded homework assignments
Final Exam	40%	The final exam is cumulative

Homework Assignments

Homework sets are project based and will be graded on a letter grade basis.

Exams

Exams will be in-person, using a blue book and will be closed notes, closed book. You will be allowed to bring in one (1) hand-written page, front and back, with any information you like.

Effort Curve

I may curve your final exam score or final course score based on overall class performance. For this curve I will take into account: (1) attendance/participation during class, (2) attendance/participation during discussion, (3) originality of submitted homework. You will be considered for a curve bonus if you engage during lecture/discussion, submit unique homework, and prepare clean notes. If I detect use of boilerplate LLM-generated code or overlap with solutions from other students that have taken this class, past or present, no effort curve will be given. The curve will never reduce your score or grade. If you receive an A on the merits (homework/exams) you will receive an A in the course.

Definition of Letter Grades

This Course uses Letter Grading: A+, A, A-, B+, B, B-, C+, C, C-, D+, D, D-, F, I, W. Grading policies follow [UCR Regulations](#). Letter grades are defined as follows:

A - Distinction, indicating exceptional achievement. Mastery of course content at the highest level of attainment that can reasonably be expected of students at a given stage of development.

B - High pass, indicating more than minimal competence but less than exceptional achievement. Strong performance demonstrating a high level of attainment for a student at a given stage of development.

C - Pass, indicating competence. Totally acceptable performance demonstrating an adequate level of attainment for a student at a given stage of development.

D - Marginal pass. This grade is used to recognize that a student's performance was marginal but clearly better than that of students who receive an F.

F - Fail. This grade is used to indicate that the student has failed the course and will not receive any credit for the course. The F grade indicates that the student's performance in the required exercises has revealed almost no understanding of the course content.

I - Incomplete. This grade is used as a temporary grade. At the discretion of the instructor, students may be given an I grade for work not completed because of a serious interruption in their work not caused by the student's negligence.

W - Withdrawal. Used on student's grade reports and transcripts to indicate all courses for which they have received official approval to drop or from which they have received official approval to withdraw after the deadlines for dropping.

Mastery Based Grading

This course will use mastery-based grading. Mastery based grading assigns letter grades "A" to "F" based on a point cutoff evaluated from the assignment rubric.

Note

Mastery based grading will never reduce your grade. If you earned a 95/100, you demonstrated mastery. The assignment difficulty is typically at a level that full credit can be achieved.

Under some circumstances I *may* allow resubmission of an assignment for you to demonstrate mastery.

Absence Policy

Consistent class attendance is the number one predictor of academic success. Therefore, class attendance is required. Excused absences include

1. Medical Conditions
2. Death in the Immediate Family
3. Military Orders
4. Legal Obligations

Written verification from a physician, lawyer, judge, or commanding officer needs to be provided for any makeup exam.

Late Homework

Late homework will receive no credit. An extension may be granted if the student can document extenuating circumstances related to an extensive period of excused absence between the date that the homework assigned and the homework due date. In addition, *religious observances* or other *unanticipated life events* may interfere with class attendance and homework submission. These *may* qualify for an extension, provided sufficient documentation is presented.

Methods of Student Instruction

- Slides: Slide deck presentations are used to deliver boilerplate content
- Lecture Notes: Handwritten lecture notes are used to deliver key derivations
- Discussion Sessions: Discussion sessions are used for one-on-one instruction

Academic Accommodations

Students are encouraged to take an active role as their own advocates. They are not required to identify themselves as having a disability; however, students are not eligible for accommodations unless they self-identify. Students are responsible for visiting the SDRC office (Student Disability Resource Center, 1228 Student Services Building, (951) 827-3861, <https://sdrc.ucr.edu/>), and obtaining and delivering letters detailing appropriate accommodations to each of their instructors in a timely manner every quarter.

Instructors are responsible for assisting in providing accommodations; students are required to make requests in an appropriate and timely manner. Ordinarily, required accommodations are described in a letter, which is given to the instructor by the student. Classroom accommodations may involve providing space for a sign language interpreter or real-time captionist, asking students to clear a path for wheelchair users, announcing need for note takers, for example. Some accommodations, such as materials in Braille, large print, or electronic format, readers, and special equipment for testing or composing papers, are provided outside of the classroom by SDRC. UCR [practices for students](#) summarized by the Dean.

Academic Rights and Responsibilities

All students, faculty, and staff are responsible for understanding and complying with the University's stated academic requirements. Students should feel free to express their thoughts and opinions in an academic forum. Assignments must be completed by the student for whom the work is assigned and without unauthorized aid of any kind.

Academic Integrity

At UCR we are committed to upholding and promoting the values of the Tartan Soul: Integrity, Accountability, Excellence, and Respect. As a student in this class, it is your responsibility to act in accordance with these values by completing all assignments in the manner described, and by informing the instructor of suspected acts of academic misconduct by your peers. By doing so, you will not only affirm your own integrity, but also the integrity of the intellectual work of this University, and the degree which it represents. Should you choose to commit academic misconduct in this class, you will be held accountable according to the policies set forth by the University, and will incur appropriate consequences both in this class and from Student Conduct and Academic Integrity Programs. For more information regarding University policy and its enforcement, please visit: <https://conduct.ucr.edu/>.

Cheating is the giving, taking, or presenting of information or material by a student that unethically or fraudulently aids oneself or another person on any work which is to be considered in the determination of a grade or the completion of academic requirements or the enhancement of that student's record or academic career. Cheating includes, but is not limited to the following actions:

1. Copying from someone else's assignment, examination, or other academic exercise.

2. Possessing, buying, selling, removing, receiving, or using, at any time or in any manner not prescribed by the faculty member, any information related to an instrument of academic evaluation.
3. Using materials, equipment, or assistance in connection with an assignment, examination, or other academic exercise which have not been authorized by the faculty member, including but not limited to, notes, calculator, or other technology.
4. Obtaining or attempting to obtain, in a dishonest manner, any material relating to a student's academic work.
5. Working with another or others in completing an assignment, examination, or other academic exercise when the faculty member has required independent and unaided action.
6. Attempting to influence or change an academic evaluation, grade, or record by unfair means.
7. Permitting another individual to substitute for one's self in an academic evaluation.
8. Marking or submitting an examination or evaluation material in a manner designed to deceive the grading system.
9. Failing to comply with a specific condition of academic integrity which has been clearly announced in a particular course.
10. Submitting, without prior permission of the faculty member, any work by a student which has at any time been submitted in identical or similar form by that student in fulfillment of any other academic requirement at any institution.
11. Submitting of material in whole or part for academic evaluation that has been prepared by another individual(s).
12. Submitting data which have been altered or contrived in such a way as to be deliberately misleading.
13. Providing false information to the University in any manner to achieve an unfair advantage, enhance one's record, or complete a requirement.

Digital Course Components

Students may be required to disclose personally identifiable information to other students in the course, via digital tools, such as email or web-postings, where relevant to the course. Examples include online discussions of class topics, and posting of student coursework. All students are expected to respect the privacy of each other by not sharing or using such information outside the course.

Digital Course Components: Digital course components include

- GitHub Classroom platform.
- Canvas LMS

Non-Discrimination Policy

It is the policy of the University of California Riverside not to engage in discrimination against or harassment of any person employed or seeking employment with the University of California on the basis of race, color, national origin, religion, sex, gender, gender expression, gender identity, pregnancy, physical or mental disability, medical condition, genetic information, ancestry, marital status, age, sexual orientation, citizenship, or service in the uniformed services. Furthermore, University policy prohibits retaliation against any member of the University community filing a complaint of discrimination or harassment. This policy is intended to be consistent with the provisions of applicable State and Federal laws and University policies.

Mental Health

As a student, you may experience a range of issues that can cause barriers to learning, such as strained relationships, alcohol/drug problems, anxiety, depression, difficulty concentrating and lack of motivation. These mental health concerns or stressful events may lead to diminished academic performance or reduce a student's ability to participate in daily activities – UCR has resources to help! If you or a friend are in distress or experiencing any of these issues, please contact 951-UCR-TALK 24-hours a day to speak to a mental health professional, or call or text the 988 Suicide & Crisis Lifeline at 988. Please refer to <https://mentalhealth.ucr.edu> for additional resources.

Basic Needs

If you or someone you know are in need of economic, food or housing support, you can find help at <https://basicneeds.ucr.edu>. You may be eligible for money to buy groceries via <https://basicneeds.ucr.edu/calfresh>. If you are in need of immediate assistance, see the Student Affairs Case Manager at <https://casemanagement.ucr.edu/> or visit the R'Pantry at <https://basicneeds.ucr.edu/rpantry>. You can find more information by clicking the Student Success Resources tab on your Canvas course menu.

Self Care

Categories of self care include biobehavioral (e.g., get plenty of sleep), affective-cognitive (e.g., watching a favorite, comforting movie), relational (e.g., talking with a friend), and spiritual (e.g., connecting with a faith community or the natural environment). In addition to the following resources, your instructor and TA are available to provide support and referrals.

University resources that may help you include:

- Counseling Center: (951) 827-5531 | [link](#)
- Wellness Center: [link](#)
- Student Health Services: (951) 827-3031 | [link](#)
- Student Special Services (for students with disabilities and veterans): [link](#)

- Ethnic and gender program offices: [link](#)
- Healing Highlanders Collegiate Recovery Program: [link](#)
- Campus Safety Escort Service: (951) 827-3772 | after 10:00 PM call UCPD at (951) 827-5222
- UCR Sexual Assault Resource Services: (951) 827-6225
- UCR Student Conduct: (951) 827-4208
- UCR Student Affairs Case Manager: (951) 827-5000
- UCR Title IX/Sexual Harassment Office: (951) 827-7070
- Office of the Ombuds: (951) 827-3213
- Campus advocate for campus members affected by interpersonal violence:

CARE (Campus Advocacy, Resources & Education) | advocate@ucr.edu
 (951) 827-6225 | [link](#)

Off-campus resources include:

- Alternatives to Domestic Violence: (800) 339-7233
- NORA, formerly Riverside Area Rape Crisis Center (24/7): (951) 686-7273
- Riverside University Health System Medical Center: (951) 486-4000
- National Sexual Assault Hotline: (800) 656-4673
- National Dating Abuse Hotline: (866) 331-9474
- National Domestic Violence Hotline: (800) 799-7233
- Family Justice Center: (951) 955-6100

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